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THE BIPOLAR JUNCTION TRANSISTOR

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1. Introduction To The BJT Transistor

The **Bipolar Junction Transistor**, or **BJT** for short, is a three-layer device made from two different types of doped semiconductor material. The bipolar transistor is able to control the flow of electric current through itself by the application of different voltage levels applied to each of its three connecting terminals.

The ability of a transistor to control the flow of current allows it to be used as either an analogue amplifier to increase or "amplify" an electronic signal, or as a digital on/off solid state switch, depending on its configuration.

Bipolar junction transistors are constructed by bonding together doped P-type and N-type semiconductor materials creating pn-junctions. As there are two pn-junctions within one transistor construction, it is therefore known as a bipolar device. Hence the name, Bipolar Junction Transistor, or BJT.

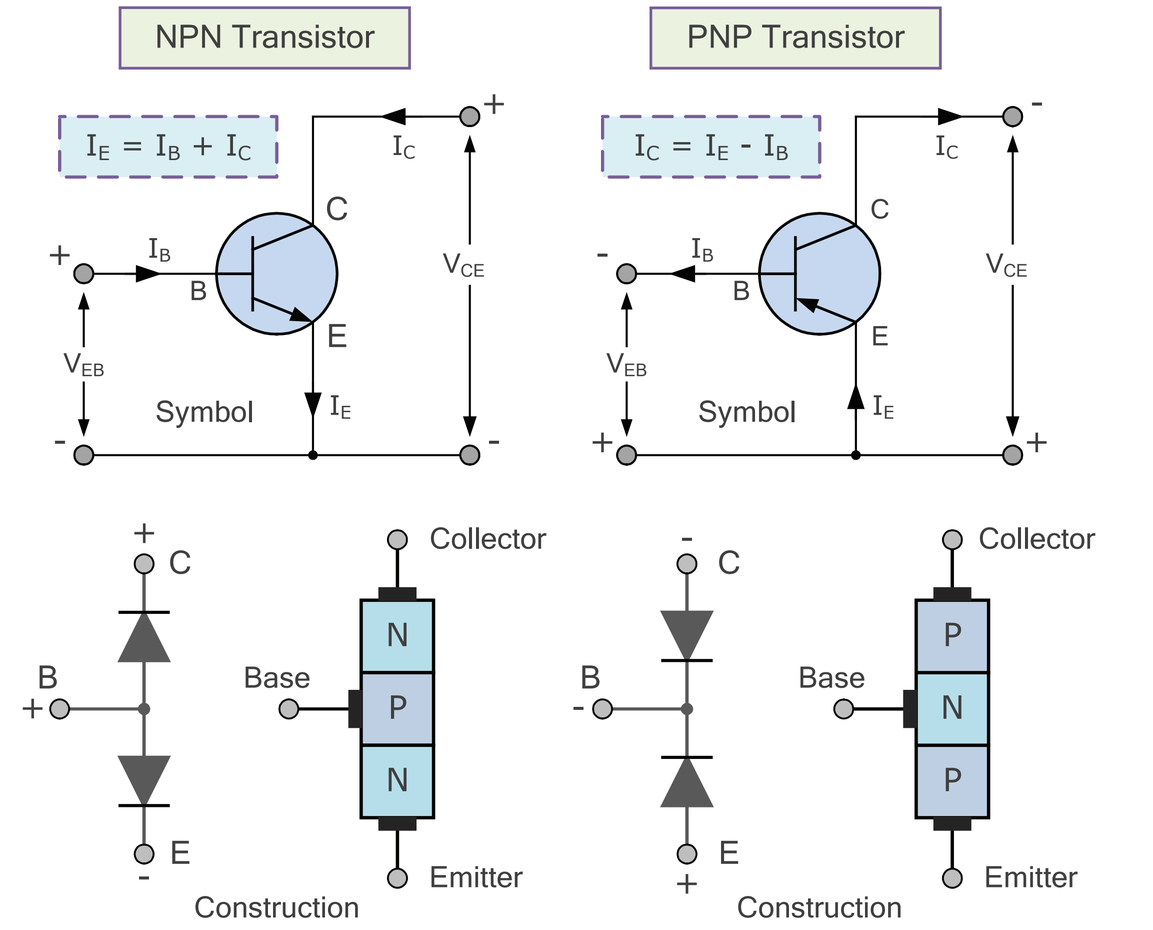
> *Transistors are 3-layer solid state semiconductors devices used in amplifier and switching circuits*

The basic construction of a Bipolar Junction Transistor is of two pn-junctions fused together to produce three distinct regions. Each of these three regions is given a name to identify it from the other two. The names of these three basic but very distinct regions as well as their connecting terminals are known and labelled as the **Emitter** (**E**), the **Base** (**B**) and the **Collector** (**C**) respectively.

A bipolar junction transistor constructed from a piece of p-type semiconductor material sandwiched between two n-type regions is called an **NPN Transistor**. Thus the base-emitter junction forms a single pn-diode, while the base-collector junction forms another pn-diode.

There also exists the complement of the NPN transistor, called the **PNP Transistor**. The PNP transistor is constructed again from a piece of n-type semiconductor material sandwiched between two p-type regions. In each case, the semiconductor layer in the centre of the doped sandwich is called the Base, (B) with the two outer layers being the Collector, (C) and the Emitter, (E) as shown in Figure 1.

Figure 1. The NPN and PNP Transistor Types



The arrow used in the transistors symbol identifies the base-emitter junction with the direction of positive current flow always pointing to the n-type material. Also note that the polarity of the applied voltages is reversed between the NPN and PNP type.

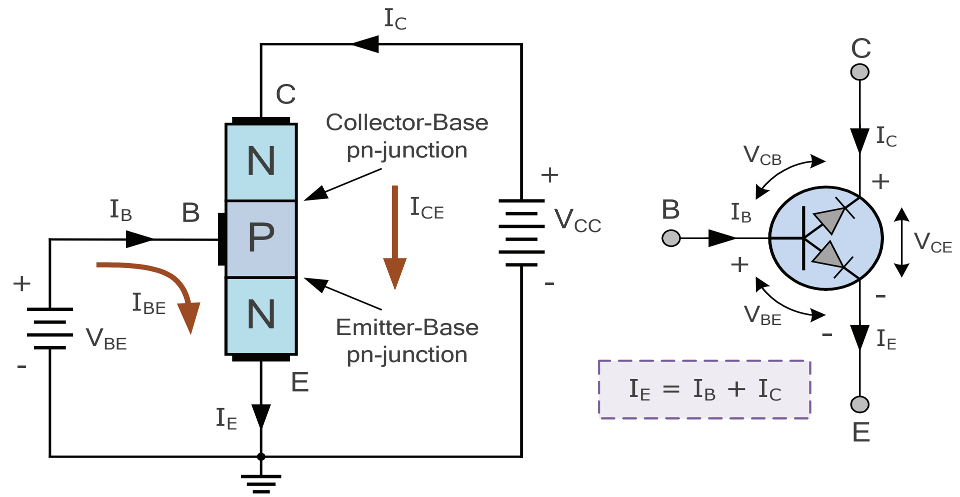
Bipolar junction transistors (BJT's) are current regulating devices. They control the flow of current through them in proportion to the amount of biasing applied to their base terminal, thus acting like a current-controlled device. The principle of operation of the two transistor types, PNP and NPN, is exactly the same the only difference being in their biasing and the polarity of the power supply connected to each type.

2. Transistor Biasing

For a transistor to operate as either an amplifier or an electronic switch, it requires the correct external voltages to be applied between all of its junctions. Although the bipolar transistors can be thought of as two diodes back-to-back, it does not however function as two individual diodes.

This because one diode, the base-emitter junction is forward biased while the other diode, the base-collector junction is reverse biased. Then for transistor action to occur, the base-emitter diode must be forward biased and the base-collector diode reverse biased by the application of the correct biasing voltages and their polarities across these two junctions as shown in Figure 2.

Figure 2. NPN Biasing Voltages



For the NPN transistor shown in Figure 2, when a supply voltage, V_{CC} is connected between the collector-emitter terminals with zero voltage applied to the base terminal, ($V_{BE} = 0$) the transistor will not conduct as the collector-base pn-junction is effectively reverse biased. Thus zero current flows into its base so it acts as an open-circuited switch.

Forward-bias is a positive voltage applied to the p-type region and a negative voltage applied to the n-type region

With Vcc still connected across the collector-emitter terminals. If a biasing voltage is now applied to the base terminal and whose value is less than 0.7 volts for a silicon transistor, or 0.3 volts for a germanium transistor compared to the voltage on the emitter, the transistor will still not conduct. This is because the base biasing voltage is too small to overcome the potential barrier voltage of the base-emitter pn-junction.

If the base biasing voltage is now increased above 0.7 volts for a silicon device, the diode junction conducts with electric current flowing into the forward biased base-emitter junction since it acts as a standard pn-junction diode supplying electrons from the n-region to the p-region.

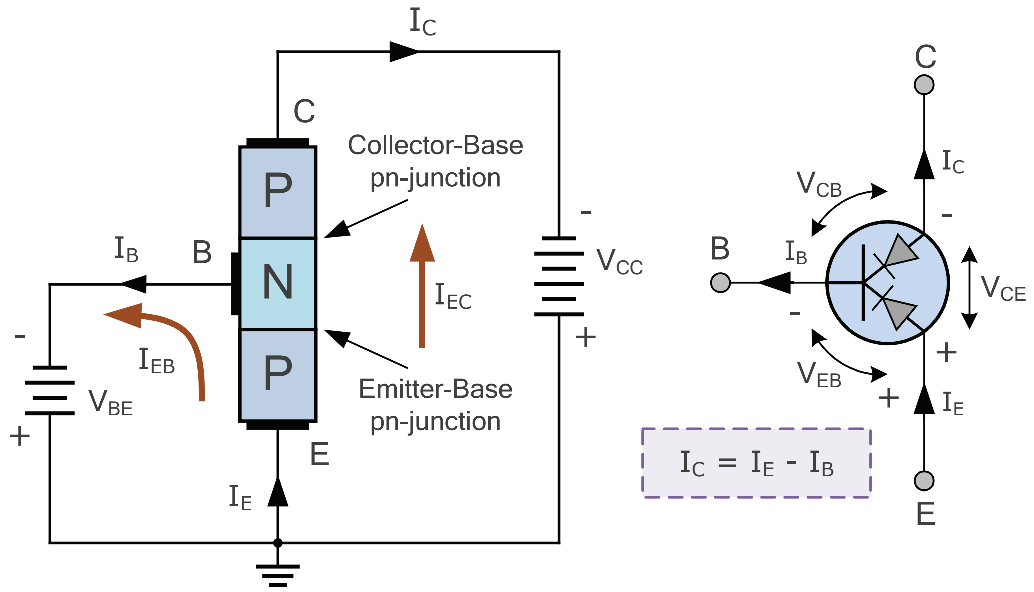
The flow of electrons into the base is now heavily influenced by the positive supply voltage, V_{CC} applied to the collector due to it being higher than the applied base voltage. Since the collector-base junction is forward biased with regards to the base terminal, this excess of electrons flow from the emitter to the collector circuit attracted by the supply voltage, V_{CC}.

Reverse-bias is a positive voltage applied to the n-type region and a negative voltage applied to the p-type region

If the base biasing voltage is increased further, the base-emitter junction becomes even more forward biased with the resistance of the conducting channel between the collector and the emitter decreasing to a very low value allowing even more current to flow between the collector and the emitter. Thus the transistor now starts to act as a closed-circuit switch.

For the PNP transistor, the voltage biases and directions of current flow are opposite to those for the NPN transistor as current will only flow from the emitter to the collector when the base terminal is forward biased by a voltage which is 0.7 volts more negative than that at the emitter. For current to flow through the emitter to the collector, both the base and the collector must be negative with respect to the emitter.

FIGURE 3. PNP BIASING VOLTAGES



3. TRANSISTOR GAIN

As the power supply, **Vcc** provides the energy required to move the current through the transistor, most of the electrons injected into the base region from the emitter are attracted towards the much larger potential present at the collector terminal.

Since the base and collector currents flow into the emitter region, the flow of current through an NPN transistor can be expressed mathematically as being:

The Δ (delta) symbol simply means a "change in" the current value

$$I_E = I_B + I_C$$

3.1 Beta (transistor current gain)

Although the base-collector junction is reverse biased, it acts as a current-controlled current source (CCCS). The amount of collector current, I_C flowing through the transistor depends heavily on the amount of base current, I_B. Thus the ratio of collector current to base current $\Delta I_C/\Delta I_B$ defines a junction transistor's DC current gain.

This DC current gain ratio is given by the Greek letter, β (*beta*) or simply h_{FE}. That is:

$$\text{Beta, } \beta_{(dc)} = I_C/I_B$$

Then **Beta** is defined as being the common emitter DC current gain. Thus the collector current of a bipolar transistor is found simply by multiplying the base current by the beta value and is expressed as: $I_C = \beta_{DC} {*} I_B$.

Most general purpose transistors, beta (β_{DC}) ranges about 25 for high power transistors to over 400 for small signal transistors, with 100 being a common standard beta value.

Unfortunately, for a given bipolar transistor beta is not always a constant value as it is strongly influenced by variations in junction temperature, collector current and can also vary between transistors of the same model number. However, there are various biasing techniques which can be used to reduce the variation of beta.

3.2 Alpha (forward current transfer ratio)

The fraction of: $\beta/(1+\beta)$ is called **Alpha**, and is given the Greek letter, α (*alpha*). Alpha is the common base current gain and is defined as the ratio between collector current and emitter current: I_C/I_E. That is:

$$\text{Alpha, } \alpha_{(dc)} = I_C/I_B$$

In general, the parameter **α** is a measure of the quality of a transistor and is very close to unity, 1. Alpha is typically in the range of 0.9 to 0.998 as the emitter current is very similar to that of the collector current. Thus $I_C \approx I_E$.

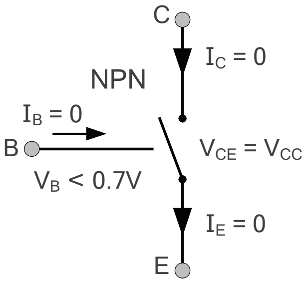
Then a bipolar transistor with a beta value of say 100, ($\beta_{(dc)}$ = 100) would correspond to an alpha value of 0.99 (α = 0.99) since : $\beta = \alpha/(1-\alpha)$.

4. Transistor Operating Regions

As a current controlled device, variations in base current, I_B cause proportional variations in the collector current, I_C. Thus the collector current is an increased version of the base terminal current showing that the bipolar transistor is a current amplifying device.

Bipolar junction transistors have the ability to operate within one of three distinct regions cut-off, active and saturation. The region of operation depends upon how the voltages on each of the three terminals relate to each other.

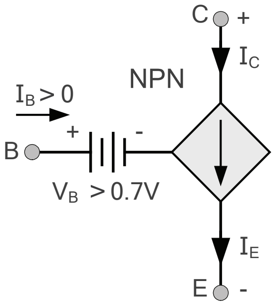
Figure 4. Transistor Cut-off Region



1. Cut-off Region – when $V_{BE} < 0.7V$, the base-emitter junction is not forward biased so no current flows into the base, $I_B = 0$. The transistor is essentially inactive and not conducting.

In cut-off mode the transistor appears as a high impedance open circuit between the collector and emitter terminals. Thus no collector current flows and the transistor is fully-off. $I_C = 0$ and $V_{CE} = V_{CC}$ = maximum voltage.

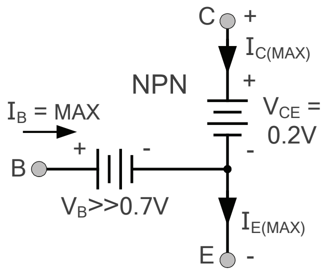
Figure 5. Transistor Active Region



2. Active Region – the base-emitter junction is forward biased and $V_{BE} \geq 0.7V$. The collector-base junction is reverse biased and the transistor operates as a fairly linear amplifier.

Collector current is proportional to the base current by the constant beta since I_C is $\beta_{DC}*I_B$. The transistors collector to emitter voltage, V_{CE} is now somewhere between fully-off and fully-on but greater than 0.2 volts. $V_{CE} > 0.2V$.

Figure 6. Transistor Saturation Region



3. Saturation Region – maximum current flows into the base with the base-emitter junction heavily forward biased since $V_{BE} >> 0.7V$.

The base current is so large that the collector-base junction becomes forward biased and the transistor is fully-on. The transistor appears as a low impedance short circuit between the collector and emitter terminals.

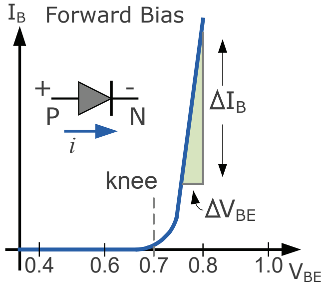
Maximum collector current flows so $I_C = I_{(sat)}$ and $V_{CE} = V_{CE(SAT)} \approx 0.2V$. As the transistor is in saturation, any increase in base current will have little or no affect on the saturated collector circuit, so the relationship of: $I_C = \beta_{DC}*I_B$ is no longer valid.

5. Bipolar Junction Transistor Characteristics Curves

Then we can see that there is a direct link between the voltage and current at the base terminal, and its collector terminal in order to drive the transistor into one of its three regions. The characteristics of a transistors operation are generally presented in the form of a set of graphs commonly referred to as a "transistors characteristics curves".

Typical *transistor characteristics curves* are a set of current-to-voltage lines on a graph showing the I–V relationship for a given bipolar transistor. I-V characteristics curves can be used to show the input and output behaviour of a device.

5.1 Junction Transistor Input Characteristics



The input characteristics of a typical junction transistor operating in common-emitter mode represent that of a forward biased diode for the base-emitter junction.

Input characteristic curve of 5.1 shows the transistors base current, I_B as a function of its base-emitter voltage, V_{BE} with the collector open. Thus I_B is plotted against V_{BE}.

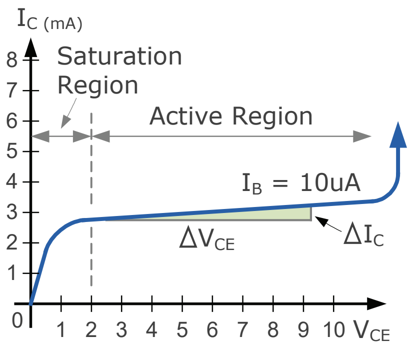
When V_{BE} is below the forward bias voltage of 0.7 volts, the barrier potential of the diodes pn-junction begins to form, but there is insufficient external base voltage to overcome it so very little current flows into the base. Thus below 0.7 volts I_B is virtually zero.

As the base-emitter voltage, V_{BE} starts to increase above 0.7 volts, the height of the potential barrier is reduced. This allows current to flow across the junction from p-type to the n-type side. The "knee" point on the characteristics occurs when $V_{BE} \approx 0.7$ volts.

As the base-emitter voltage increases further, the current flowing through the junction also increases at a much larger rate showing that the pn-junction exhibits an exponential current–voltage characteristic for a normal silicon diode.

5.2 Junction Transistor Output Characteristics

The output characteristic for a bipolar junction transistor is its collector current, I_C as a function of collector-emitter voltage, V_{CE} for a given amount of base current, I_B. It is these two quantities of output current, I_C and output voltage, V_{CE} that represents the output characteristics of the transistor when connected in a common emitter configuration.



The characteristics curve of 5.2, shows how an NPN transistors output responds to a fixed value of base current. In this example, $I_B = 10uA$.

As the collector-emitter voltage V_{CE} is increased for a given base bias current. The transistor starts to conduct allowing collector current to flow. Collector current, I_C increases steadily in response to V_{CE} up to the switching "knee" point of about 2 volts.

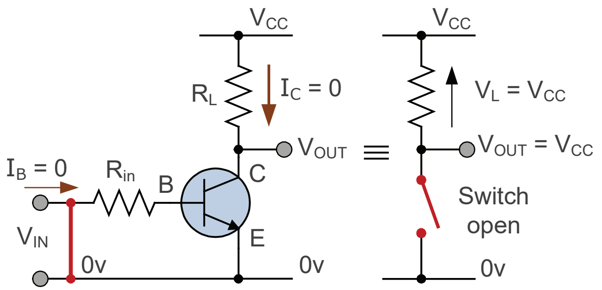
The characteristic curve then flattens out, as above this value, any large change in the collector-emitter voltage results in very little change of collector current for a given base bias current. Thus this curve can also be referred to as the transistors constant current characteristic since a fixed base current will regulate the amount of collector current at a value determined by beta.

Changing the value of the base biasing current I_B, we can create a different set of curves to plot I_C against V_{CE} for different values of I_B from zero to saturation. These set or family of collector current curves will tell us a great deal about the operation of the transistor.

6. The Bipolar Transistor as a Switch

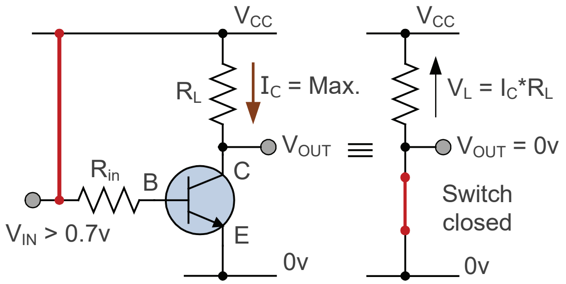
A bipolar junction transistor can be used as an amplifier or as an electronic switch depending on its operating region. When used as an ON/OFF electronic switch, the transistor is biased so that it operates between its cut-off region (switched off) and its saturation region (switched on).

Figure 7. Open Circuit (Switch) Condition (cut-off)



- The base terminal is grounded (0v)
- Base-emitter voltage $V_{BE} < 0.7$ volts
- Base-emitter junction is reverse biased
- Transistor is "fully-OFF" (cut-off region)
- No collector current flows ($I_C = 0$)
- $V_{CE} = V_{CC}$ no voltage drop across R_L
- $V_{OUT} = V_{CE} = V_{CC} = 1$ (logical one)
- The output is opposite to the input
- Operates as an "open switch"

Figure 8. Closed Circuit (Switch) Condition (saturation)

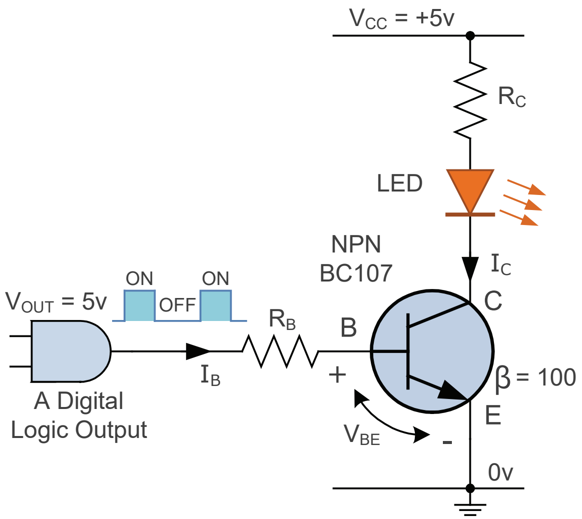


- The base terminal is connected to V_{CC}
- Base-emitter voltage $V_{BE} >> 0.7$ volts
- Base-emitter junction is forward biased
- Transistor "fully-ON" (saturation region)
- Max collector current flows: $I_C = V_{CC}/R_L$
- $V_{CE} < 0.2v$ (ideal saturation)
- $V_{OUT} = V_{CE} = 0$ (logical zero)
- The output is opposite to the input
- Operates as a "closed switch"

7. Transistor Switch Application

Transistor switches have many applications from switching digital circuits to high power loads. By driving the transistor hard into its saturation region, nearly all the power supply voltage V_{CC} is dropped across the connected load allowing relatively large collector load currents to flow through the transistor without causing it to overheat as V_{CE} is generally less than 0.5 volts.

Figure 9. Transistor Switching Circuit



Let's assume we want to use a transistor as a switch to operate an orange coloured Light Emitting Diode (LED) indicator lamp using a Digital or Arduino logic circuit as shown in Figure 9.

Let's also assume that the output from the logic circuit is 0 (LOW) and +5 volts (HIGH), and the supply voltage, V_{CC} is +5V.

The beta (β) value of the NPN BC107 transistor is 100, V_{CE} when fully-on is 0.2 volts, and the forward voltage drop, V_F of the LED is 2 volts when illuminated.

What would be the values of R_B and R_C if the LED draws a current or 12mA when illuminated.

Since V_{CC} = 5v, applying Kirchhoff's voltage law (KVL) to the collector circuit:

$$V_{CC} - V_{RC} - V_{LED(ON)} - V_{CE(SAT)} = 0, \quad \text{thus: } 5 - V_{RC} - 2 - 0.2 = 0, \quad \text{therefore: } V_{RC} = 2.8 \text{ volts}$$

$$R_C = \frac{V_{RC}}{I_C} = \frac{2.8}{0.012} = 233\,\Omega$$

or 232Ω to the nearest preferred value. This LED series resistor, R_C will dissipate under 33mW ($I^2 * R$) when the LED is illuminated. So a 1/4 watt resistor is more than sufficient.

The DC current gain value of the transistor is 100. Thus $I_B = I_C/\beta_{DC} = 0.012/100 = 120\text{uA}$

The output from the digital circuit switches between 0 and 5 volts, so applying Kirchhoff's voltage law (KVL) again to the base circuit:

$$V_{DIGITAL} - V_{RB} - V_{BE} = 0, \quad \text{thus: } 5 - V_{RB} - 0.7 = 0, \quad \text{therefore } V_{RB} = 4.3 \text{ volts}$$

$$R_B = \frac{V_{RB}}{I_B} = \frac{4.3}{0.00012} = 35834\,\Omega \text{ or } 35.8\text{k}\Omega$$

or 36kΩ to the nearest preferred value. Base resistor, R_B will dissipate less than 520uW, so again a standard 1/4 watt resistor is more than sufficient. Reducing R_B will increase I_B and drive the transistor harder into its saturation region.

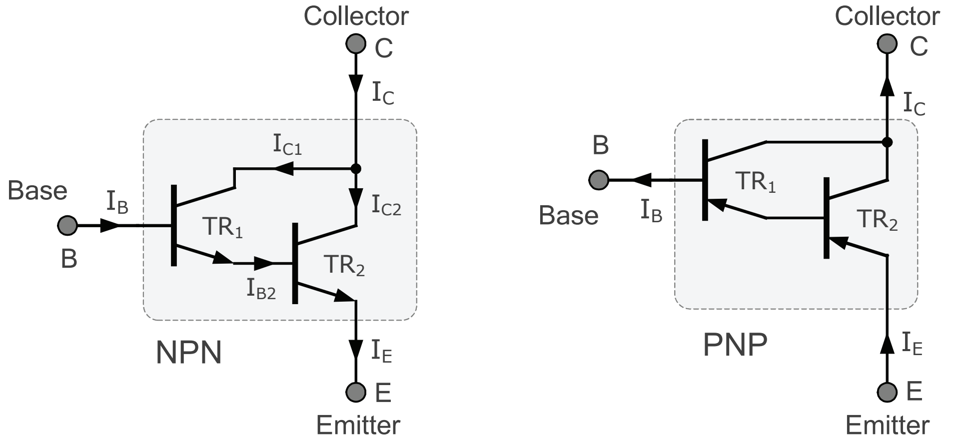
8. Darlington Transistor Configuration

We have seen previously that when operated as a switch, the ratio of collector current to base current ($\Delta I_C/\Delta I_B$) is known as the current gain or beta value of the junction transistor. Typical values of β for a standard bipolar transistor are in the range of 50 to 400 and varies even between transistors of the same part number.

In some cases where the current gain of a single transistor is too low to directly drive a given load, for example a power transistor. One way to increase the overall current gain is to use a two interconnected devices acting as one single transistor switch.

A dual transistor configuration, also known as a "Darlington pair", consist of two NPN or PNP transistors connected together back-to-back, so that the emitter current of the first transistor TR_1 becomes the base drive current of the second transistor TR_2. In other words, the first transistor (TR_1) is connected as an *emitter follower* and the second transistor (TR_2) is connected as a common emitter amplifier as shown in Figure 10.

Figure 10. Basic Darlington Transistor Configuration



Using the NPN Darlington pair as the example, the collectors of two transistors are connected together, and the emitter of TR_1 drives the base of TR_2. This configuration achieves β multiplication since the base current, I_{B2} is equal to transistor TR_1 emitter current, I_{E1} as the emitter of TR_1 is connected to the base of TR_2. Therefore:

$$I_C = I_{C1} + I_{C2}$$

$$I_C = (\beta_1 \times I_B) + (\beta_2 \times I_{B2})$$

$$I_C = [\beta_1 + (\beta_1 \times \beta_2) + \beta_2] \times I_B$$

Where: β_1 and β_2 are the gains of the individual transistors.

This means that the overall current gain, β is given as the current amplified by the first transistor is amplified further by the second transistor as the two current gains multiply.

Thus two bipolar junction transistors connected together can make one single Darlington configuration with a very high beta value. For example, two interconnected transistors with a beta of 100 and 50 respectively, would give a total beta value of 5000. That is I_C can be 5000 times greater than I_B. Effectively creating a super-beta transistor.

If two identical bipolar transistors are used to make one single Darlington configuration. If beta β_1 is equal to β_2, then the overall current gain is given as:

$$\text{If} \quad \beta_1 = \beta_2$$

$$\text{and} \quad I_C = [\beta_1 + (\beta_1 \times \beta_2) + \beta_2] \times I_B$$

$$\text{then:} \quad I_C = (\beta^2 + 2\beta) \times I_B$$

Generally, the value of β^2 is much, much greater than that of 2β, in which case the 2β part of the equation can be ignored to simplify the maths. Thus: $I_C = \beta^2 \times I_B$ which shows that a Darlington pair behaves just like a single transistor but with a very high DC current gain.

The advantage of using a double transistor arrangement such as this, is that Darlington transistor pairs can have very large current gains allowing collector current loads of several amperes. For example: the NPN TIP120 or the TIP30xx series and their PNP equivalents TIP125 or TIP29xx series.

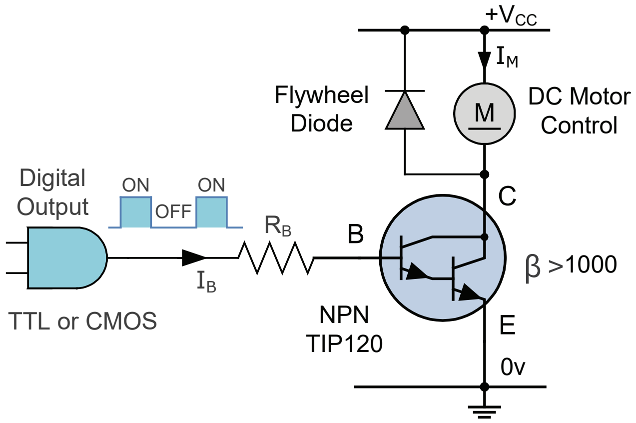
Thus Darlington transistor pairs are much more sensitive to input signals as only a very small base current is required to switch a much larger load current with the typical gain of a Darlington configuration being over 1,000. Whereas normally a single transistor stage produces a DC current gain of only 50 to about 400.

> *The Darlington configuration is simply two back-to-back bipolar transistors*

Then we can see that a Darlington transistor pair with a DC current gain of 1,000:1 could switch an output load current of 1 ampere in the collector-emitter circuit with an input base current of just 1mA.

Since the switching characteristics of the Darlington pair are basically the same as those for a single transistor emitter follower circuit, they can both be used in similar applications. Making Darlington configurations, either as an NPN pair or PNP pair, ideal for high power switching applications such as interfacing relays, lamps, solenoids and DC motors to low power microcontroller, Arduino's or TTL & CMOS logic gates.

Figure 11. Darlington Transistor Switch



Here in the circuit of Figure 11. the base of the Darlington transistor is sufficiently sensitive to respond to any small input current from a switch or directly from a TTL or 5V CMOS logic gate. The maximum collector current $I_{C(max)}$ for any Darlington pair is the same as that for the main switching transistor.

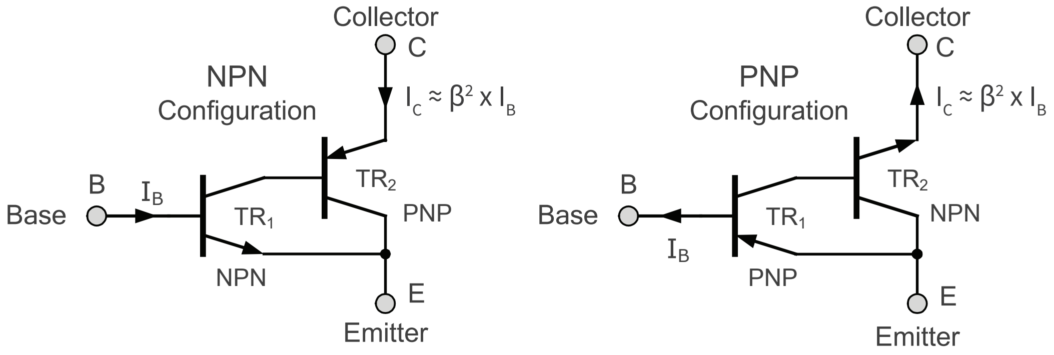
One of the main disadvantages of a Darlington transistor pair is the minimum voltage drop between the base and emitter when fully saturated. Unlike a single transistor switch which has a saturated voltage drop ($V_{CE(SAT)}$) of between 0.2v and 0.6v when fully-ON. A Darlington device can have over twice the base-emitter voltage drop as the base-emitter voltage is the sum of the two individual transistor diode junctions.

Thus the collector-emitter saturation voltage, $V_{CE(SAT)}$ of a Darlington's interconnected arrangement is much higher than that of a single bipolar transistor. To overcome this issue the Sziklai Darlington Pair was developed.

9. The Sziklai Darlington Pair

The Sziklai Darlington Pair, named after its Hungarian inventor George Sziklai, is a complementary or compound Darlington configuration which consists of separate NPN and PNP complementary transistors connected together to overcome some of the disadvantages of using two NPN or two PNP devices.

Figure 12. Sziklai Darlington Pair



The Sziklai Darlington pair configuration of Figure 12. contains two opposite polarity bipolar junction transistors within its configuration. The cascaded combination of an NPN and a PNP transistor has the advantage that the Sziklai pair performs the same basic function of a Darlington pair except that it only requires 0.6v for it to turn-ON.

Similar to the standard Darlington configuration, the DC current gain is equal to β^2 for equally matched transistors or as before, is given by the product of the two current gains for unmatched individual transistors.

10. The Bipolar Transistor as an Amplifier

As well as being used as a solid state switch to turn connected loads "ON" or "OFF" by controlling their base signal, one of the main applications of bipolar junction transistors is in signal amplification. Because as we have seen, a small current applied to the base terminal can control the flow of a much larger current in the collector terminal from a DC power source.

> *Bipolar transistor amplifiers operate entirely in their active region due to fixed or self-biasing methods*

Biasing the transistor in such a way as to operate it in its active region. The transistor will amplify any small AC signal applied to its base with the emitter grounded. Thus the basic transistor switch circuit can also be used as an amplifying device.

One such commonly used configuration of an NPN transistor is that of the "Common-emitter Amplifier" configuration, producing what is generally called a "Class-A amplifier" circuit. But there are many other kinds of BJT amplifier circuit configurations, each with its own special features, advantages and applications.

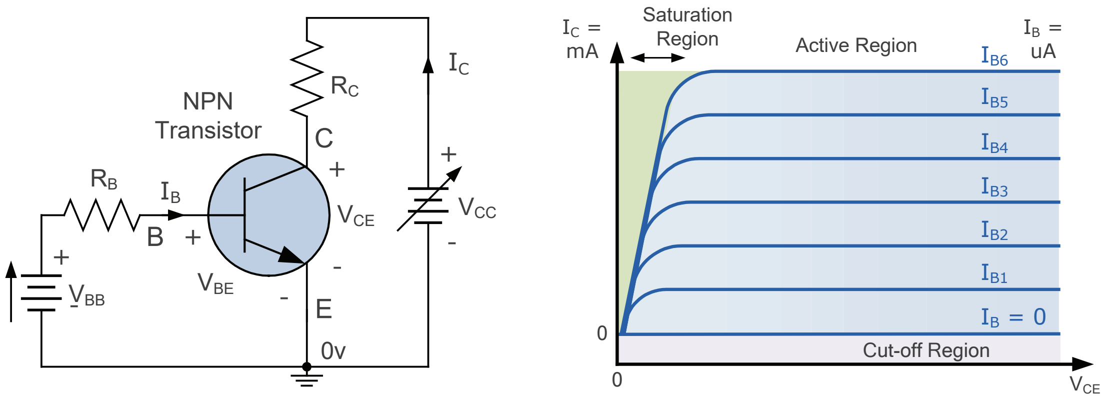
10.1 The Common Emitter Amplifier

Biasing a bipolar junction transistor means finding the required value of collector-emitter voltage V_{CE} and collector current I_C required to continually forward bias the base-emitter junction halfway between its cut-off and saturation regions.

Transistor biasing allows the amplifier circuit to accurately reproduce the positive and negative halves of any ac input signal, thus producing an undistorted output voltage waveform. The values of V_{CE} and I_C define the amplifiers quiescent operating point or Q-point which only occurs when the transistor is correctly biased into its active region.

We recall previously that input and output characteristics curves can be plotted for any transistor and that the output curve is created using a fixed base current value for varying values of collector-emitter voltage, V_{CE}. Thus we can create a set or family of output characteristics curves for different values of base current, I_B as shown in Figure 13.

FIGURE 13. DETERMINING THE I-V CHARACTERISTICS OF A BJT



Having created a set of characteristics curves for a particular transistor, we can determine its quiescent operating, or Q-point. This defines its operating region to give an undistorted amplified output by setting the required biasing current and voltage conditions.

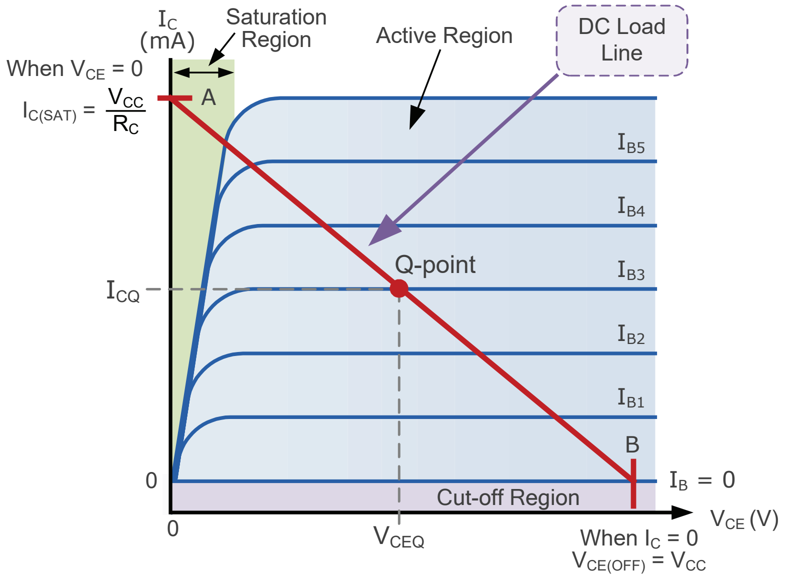
The Q-point can be fixed anywhere within the active region but to ensure that both the positive and negative half cycles of the ac input waveform are correctly amplified, the Q-point is generally centred around the middle of the previous characteristic curves to provide the maximum symmetrical output voltage swing.

10.2 DC Load Line Construction

When operated as a solid state switch, the transistor is biased between its cut-off and saturation regions. In its cut-off region, $I_C = 0$ and V_{CE} is at maximum as $V_{CE(OFF)} = V_{CC}$. Likewise when it is in its saturation region, I_C is maximum and $V_{CE} = 0.2v$ (zero volts for ideal condition). Collector current I_C is limited only by the collector load resistance, R_L.

A DC load line can be plotted on the characteristic curves between $V_{CE(OFF)} = V_{CC}$ (point "B") on the x-axis, and $I_{C(SAT)} = V_{CC} / R_L$ (point "A") on the y-axis as shown in Figure 14.

FIGURE 14. DC LOAD LINE



Having determined the position of the Q-point on the centre of the load line, the quiescent currents and voltages, I_{CQ} and V_{CEQ} respectively can be found. As the Q-point is centred on the load line between points A and B, and with no ac input signal applied, I_{CQ} will be equal to half $I_{C(SAT)}$ and V_{CEQ} will be equal to half V_{CC}.

Thus centring the Q-point on the DC load line allows for optimum ac operation of the bipolar transistor amplifier when connected in its common emitter configuration allowing the relationship of: $I_C = \beta_{DC} * I_B$ to be valid once again.

The position of the Q-point on a DC load line determines the maximum undistorted output signal for a given input signal

However, the Q-point is greatly affected by changes in a transistors beta value. Changes in operating temperature or the replacement of the transistor for another will cause changes in beta. Resulting in the movement of the Q-point along the load line affecting its DC operating point. Biasing the transistors base with a fixed voltage value prevents, or greatly limits the movement of the Q-point thereby increasing stability of the amplifier circuit.

10.3 The Base Biasing of a Junction Transistor

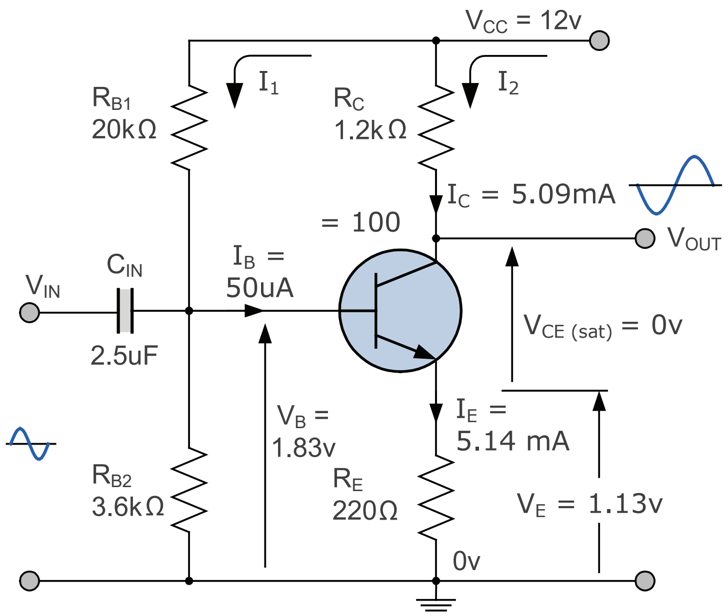
There are many different ways to bias the base terminal a bipolar junction transistor, such as: *fixed current bias*, *emitter feedback biasing*, *collector feedback biasing*, *dual supply biasing*, etc. but the most common form of biasing the base of a transistor is by using a voltage divider network.

Consider the common emitter (CE) transistor amplifier of Figure 15. which has a voltage divider biasing network of R_{B1} and R_{B2}, plus an emitter resistance, R_E. Remember that the common emitter amplifier has a load resistor in its collector circuit. The collector current through this resistance produces the voltage output of the amplifier.

1). The open-circuit base biasing voltage divider value:

$$V_B = \left(\frac{R_{B2}}{R_{B1}+R_{B2}}\right)V_{CC} = \left(\frac{3k6}{20k+3k6}\right)\times 12 = 1.83V$$

FIGURE 15. DC ANALYSIS OF THE AMPLIFIER CIRCUIT



2). Find the emitter voltage, V_E

$$V_E = V_B - V_{BE} = 1.83 - 0.7 = 1.13 \text{ Volts}$$

3). Find the emitter current, I_E

$$I_E = V_E/R_E = 1.13/220 = 5.14\text{mA}$$

4). Determine the collector (load) current, I_C

$$I_C = \left(\frac{\beta}{\beta+1}\right)I_E = \left(\frac{100}{100+1}\right)\times 0.00514 = 5.09\text{mA}$$

Note that the difference between I_C and I_E will be the base current I_B as: $I_B = I_E - I_C$

5). The quiescent base current, I_B

$$I_B = I_E - I_C = 0.00514 - 0.00509 = 50uA$$

6). The voltage drop across load resistor, R_C

$$V_{RC} = I_C \times R_C = 0.00509 \times 1200 = 6.1 \text{ Volts}$$

7). The collector-emitter voltage, V_{CE}

$$V_{CE} = V_{CC} - V_{RC} - V_E = 12 - 6.1 - 1.13 = 4.77 \text{ Volts}$$

8). Biased DC output voltage, $V_{OUT(DC)}$

$$V_{OUT} = V_{CC} - V_{RC} = 12 - 6.1 = 5.9 \text{ Volts}$$

Thus the Q-point sits halfway along the load line and is inside the Active region of the characteristics as shown in Figure 16.

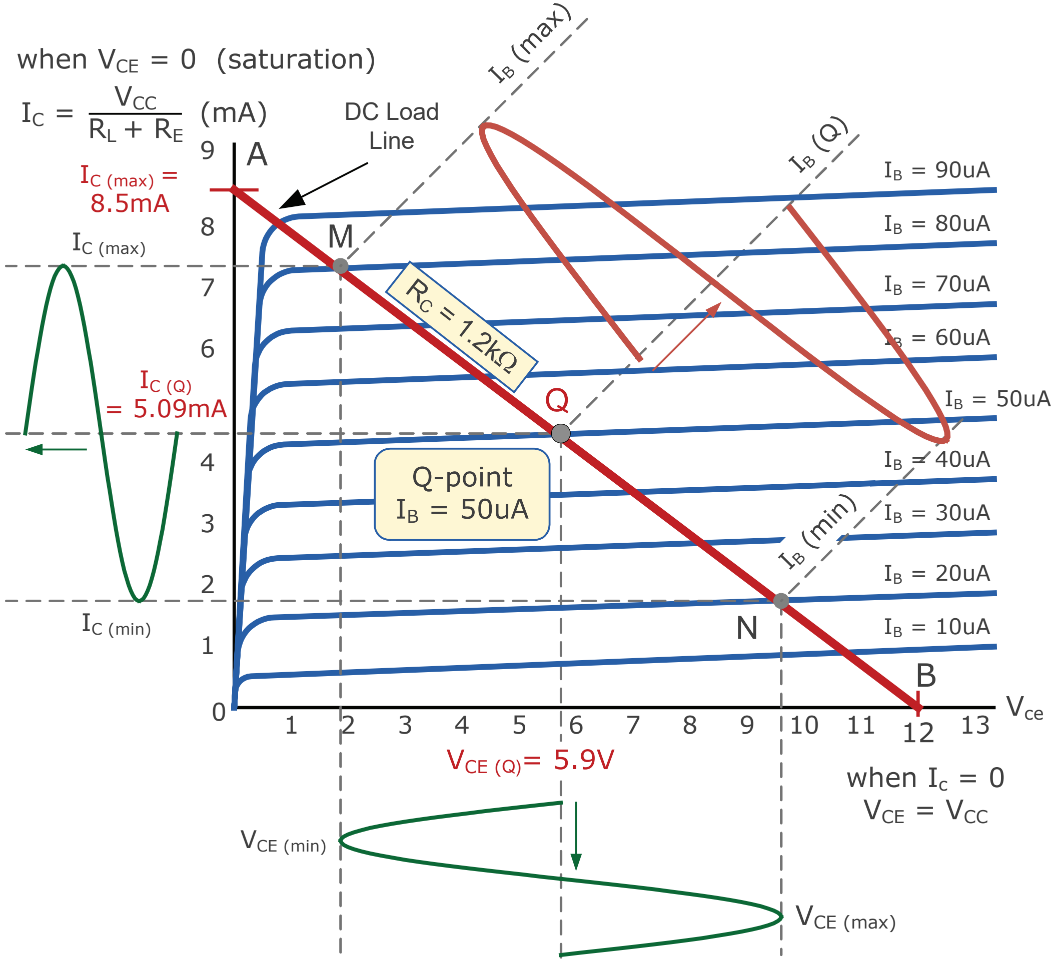
Having calculated the DC biasing network, we can turn our attention to the small signal analysis of the circuit when an ac input signal is applied. The input signals path to ground is through the forward-biased base-emitter junction. The dynamic resistance of this junction is given as a resistor, **r'e**.

As the dynamic resistance of a forward-biased diode junction is equal to **25 mV** divided by the current flowing through the diode junction, the formula used for calculating the dynamic resistance, **r'e** of the forward-biased base/emitter junction is **r'e = 25 mV/** I_E. Where I_E is the emitter current. Thus the transistors dynamic leg resistance **r'e** is calculated as being:

8). The dynamic emitter leg resistance, r'e

$$r'e = 25mV / I_E = 0.025/0.00514 = 4.9\ \Omega$$

FIGURE 16. AMPLIFIER CHARACTERISTICS CURVES



The voltage gain of the transistor amplifier is given as: $Av = V_{OUT}/V_{IN}$ where $V_{OUT} = I_C \times R_C$ and $V_{IN} = I_E \times (R_E + r'e)$. Since I_E is approximately equal to I_C ($I_E \approx I_C$) the two current values effectively cancel each other out giving the voltage gain of the circuit as:

$$A_V = \frac{V_{OUT}}{V_{IN}} = -\left(\frac{R_C}{R_E + r'e}\right) = -\left(\frac{1200}{220 + 4.9}\right) = -5.36$$

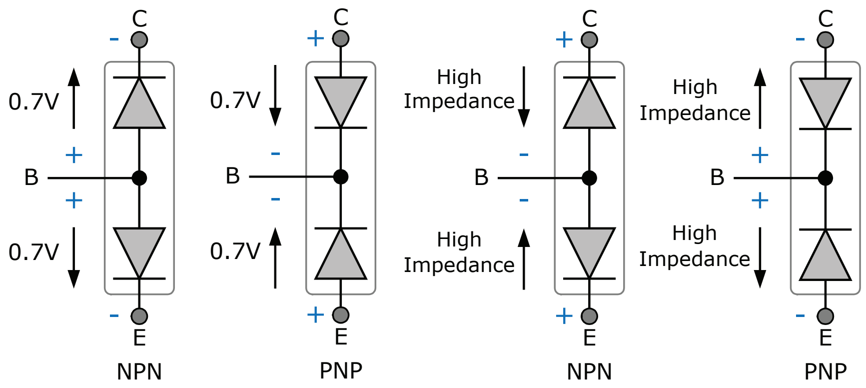
Thus for an input signal voltage of 0.5Vpk, the output voltage will be: -5.36*0.5 = -2.68Vpk.

11. Testing a Bipolar Junction Transistor

Bipolar transistors are three-layer two-junction devices so can be viewed more simply as two series connected diodes with a common connection. By assuming that a bipolar transistor is nothing more than two diodes connected at the base and having separate emitter and collector leads, transistor can be identified using just a basic multimeter which has a diode test function or a low DC voltage setting.

To determine the type of transistor, the multimeter leads are connected in different combinations across the three connecting terminals identifying the two diode junctions of Figure 17. Thus forward biasing a pn-junction will have a low resistance while reverse biasing the pn-junction will give an infinite resistance.

Figure 17. Bipolar Transistor Two-diode Model for Testing



For an NPN transistor, the red (positive) multimeter lead is connected to the base (B) terminal, and the black (negative) lead will be connected to either the emitter (E) or the collector (C) terminal. In both directions, base-to-collector and base-to-emitter should read between 0.6 and 0.7 volts (for silicon) as they are forward biased. If either direction reads high impedance on the multimeter, then the chances are it's a PNP transistor in reverse bias.

For an PNP transistor, the black (negative) lead will be connected to the base, and the red (positive) lead to either the emitter or collector. Again in both directions, base-to-collector and base-to-emitter should read between 0.6 and 0.7 volts as they are forward biased. If either direction reads high impedance on the multimeter, then the chances are it's an NPN transistor in reverse bias.

This type of test can quickly show what type of transistor you have an NPN or a PNP, but make sure to note the results and polarise the test leads accordingly. Another very simple and effective way to identify whether the device is an NPN or PNP transistor, is to look up the device part number on the internet for its datasheet and operating parameters.

End of this Bipolar Junction Transistor eBook

With the completion of this Bipolar Junction Transistor eBook you should have gained a basic understanding and knowledge of bipolar transistors. The information provided here should give you a firm foundation for continuing your study of electronics and electrical engineering. In ebook 14 you will learn about the Field Effect Transistor, or FET.

For more information about any of the topics covered here please visit our website at:

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